

COSE UTILI XOR: $A \oplus B = A\bar{B} + \bar{A}B$

LEGGI DI DE MORGAN: $\overline{AB} = \bar{A} + \bar{B}$ e $\overline{A+B} = \bar{A} \cdot \bar{B}$, $\bar{\bar{A}} = A$, $A\bar{A}$ NON SI VERIFICA MAI

DERIVATA DI $1/x \rightarrow -1/x^2$

$E = \frac{R_{N1}}{R_{P1}}$, $\alpha = \frac{C_P}{S_N}$, $R_N = R_{N1} \cdot S$ e $S = \frac{W}{L}$

SERIE MOSTET: $S_{EQ} = \frac{S_1 \cdot S_2}{S_1 + S_2}$ SE $S_1 = S_2 = S \rightarrow S_{EQ} = \frac{S}{2}$ (TRA 2 MOS)
 $S_{EQ} = \frac{S}{3}$ (TRA 3 MOS)

METTERE CORRENTE

PARALLELO MOSTET: $S_{EQ} = S_1 + S_2$

METTERE CORRENTE

$$F(r) = \frac{1}{1-r} \left[\frac{r}{1-r} + \frac{1}{2} \ln(1g - 20V) \right]$$

N-MOSTET: TRIODO: $V_{DS} \leq V_{GS} - V_T \rightarrow I_{DS} = R_N [(V_{GS} - V_T)V_{DS} - \frac{1}{2}V_{DS}^2]$

⚠️ DRAIN SEMPRE SATURAZIONE: $V_{DS} > V_{GS} - V_T \rightarrow I_{DS} = \frac{R_N}{2} (V_{GS} - V_T)^2 = g_D$ (LOW INVERT)
 MASSIMA OFF: $V_{GS} < V_T \rightarrow I_{DS} = 0$

TIPICAMENTE DRAIN ALTO $\rightarrow$ SATURAZIONE, DRAIN BASSO $\rightarrow$ TRIODO

P-MOSTET: TRIODO $V_{DS} > V_{GS} - V_T \rightarrow I_{SD} = R_P [(V_{GS} - V_T)V_{DS} - \frac{1}{2}V_{DS}^2]$

SATURAZIONE: $V_{DS} \leq V_{GS} - V_T \rightarrow I_{SD} = \frac{R_P}{2} (V_{GS} - V_T)^2$

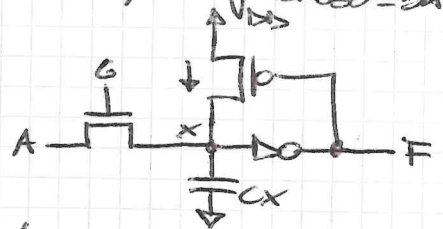
⚠️ V_{GS} , V_{DS} e $(V_{GS} - V_T)$ SONO TENS. NEGATIVE, POSSO GIRARE I DIF.

OFF: $V_{GS} > V_T$ ⚠️ DRAIN SEMPRE ALLA TENS. MINIMA / SOLIT: DRAIN ALTO = TRIODO BASSO = SAT.

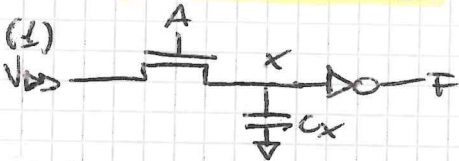
MOS DI CANALE: $I_{DS} (1 + \lambda V_{DS})$ o $I_{SD} (1 + \lambda V_{SD})$

EFF BODY: N-MOS: $V_T = V_{T0} + \gamma (\dots)$

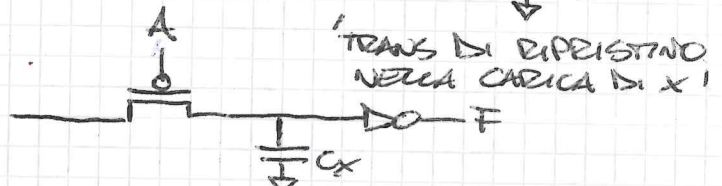
P-MOS: $V_T = V_{T0} - \gamma (\sqrt{2\phi_F + V_{GS}} - \sqrt{2\phi_F})$



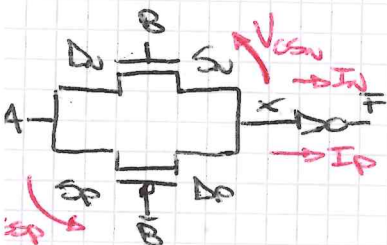
PASS TRANSISTOR



'1 DEBOLE $\rightarrow F = V_{DD} - V_T$ QUANDO CARICO CX



'0 DEBOLE $\rightarrow F = V_T$ QUANDO SCARICO CX



'PASS TRANS. COMPLEM'

$$R_{EQ} = \frac{2V_{DD}}{R_N (V_{DD} - V_{TN0})^2 + R_P (V_{DD} - V_{TP0})^2}$$

$$\epsilon_{V,X} \approx 2,3 R_{EQ} C_X$$



$$T_N = R_{EQ} \cdot C \frac{N(N-1)}{2} \rightarrow \epsilon_{V,X} \approx 2,3 T_N$$



1) TRANSISTOR DA SPRENDERE IN BLOCCHI DA M

$(\frac{N}{m})$ BLOCCHI DA M TRANS + $(\frac{N}{m} - 1)$ BUFFER

$$\epsilon_P \approx 2,3 T_m \frac{N}{m} + (\frac{N}{m} - 1) T_{BUFF} = 2,3 R_{EQ} C \frac{N(m-1)}{2} + (\frac{N}{m} - 1) T_{BUFF}$$

$$f_{OPT} \rightarrow \frac{\partial \epsilon_P}{\partial m} = 0 \rightarrow m_{OPT} = \sqrt{\frac{2 T_{BUFF}}{\epsilon_P}}$$

LOGICHE STATICHE

$FU(I_1, I_2, \dots, I_N) = F(I_1, I_2, \dots, I_N) \rightarrow$ SE PU CONDUCE, $F=1$
 $PD(I_1, I_2, \dots, I_N) = \overline{F(I_1, I_2, \dots, I_N)} \rightarrow$ SE PD CONDUCE, $F=0$

SOMMA: // DI MOSTRET, PRODOTTO: SERIE DI MOSTRET

$t_{\phi} = \frac{2C_L}{\beta' S_N V_{DS}} F\left(\frac{V_I}{V_{DS}}\right)$, $t_r = \frac{2C_L}{\beta' S_P V_{DS}} F\left(\frac{V_I}{V_{DS}}\right) \rightarrow t_{\phi} = t_r$ SE $\beta' S_N = \beta' S_P$
 $\hookrightarrow S_P = S_N$

PER AVERE STESSO RT NELL'INVERTER

$S_N = S_{INV}$, $S_P = ES_N$

CAPACITA' DI INGRESSO (SE OGNI INGR E' COLL. A UN N-MOS NEL PD E A UN P-MOS NEL PULL-UP) O COME UN INV)

$C_{IN} = (S_N + S_P) C_{mg} \rightarrow C_{mg} = C_{ox} L_{min}^2 + 2C_{GS} L_{min}$

OCCUPAZIONE D'AREA:

$S_N = \frac{W_N}{L_N} \rightarrow W_N = S_N L_N \rightarrow A = W_N L_N$ (STESSO PER P-MOS)

E' L'AREA DI UN TRANS, SE HO N TRANS, $A_{TOT} = N \cdot A$

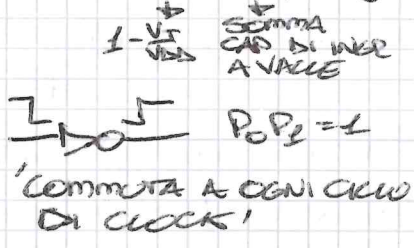
SELF LOADING (FAN IN = m, FAN OUT = k, CWR CAP DI INTERCONN)

$C_L = K C_{IN} + C_W + S_N (1+m) C_{PI} \rightarrow C_{PI} = L_{min} [2C_{GS} + \frac{1}{2} L_{min} (C_{ox} P_{DS} + K_{ox} L_{ox} C_{GS})]$

POTENZA DINAMICA

$P_{DIN} = C_L V_{DS}^2 f_{0 \rightarrow 1}$

$V_{DS}^2 \rightarrow$ SWINGS BITENS
 $f_{0 \rightarrow 1} = f_{CLK} \cdot P_{0 \rightarrow 1}$
 $P_{0 \rightarrow 1} = P_0 \cdot P_1$



SE LE PROB DI '0' e '1' IN INGRESSO SONO UGUALI

$P_0 = \frac{N_0}{2N}$ e $P_1 = \frac{N_1}{2N}$ (N_0 = NUM DI '0' IN USCITA NELLA TAB DIVERITA'
 $N_1 = \dots$)

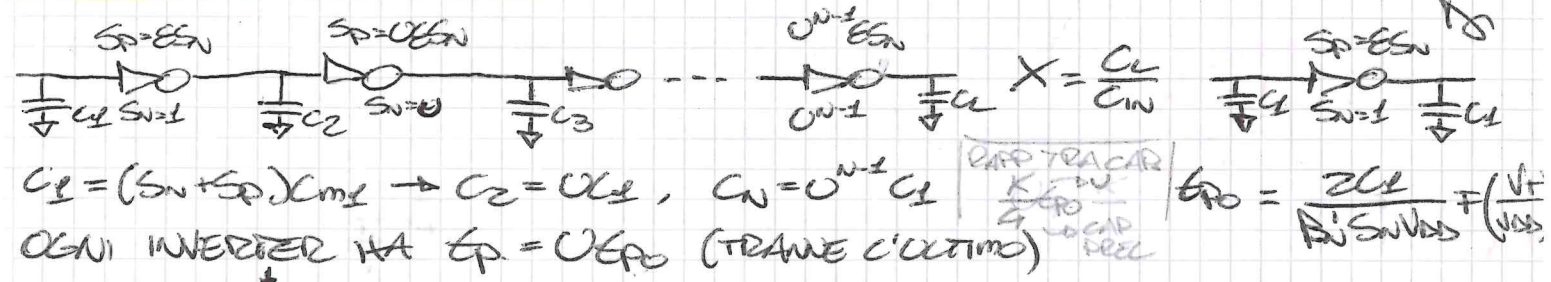
SE LE PROB NON SONO LE STESS: $P_A = \text{PROB } A=1$, $P_B = \text{PROB } B=1$

E IN BASE ALLA TAB DIVERITA' VALUTO QUANDO $F=1$ O $F=0$

$(P_{A=0} = (1 - P_A), P_{B=0} = (1 - P_B))$ AD ES: SE $P_{IN=0} = \frac{2}{3} \Rightarrow P_{IN=1} = (1 - P_{IN=0})$

CAECATA DI BUTTER

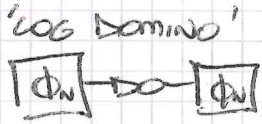
$P_0 = P_A=0 \cdot P_B=0 \cdot P_C=0 \rightarrow$ NELLA TAB HO VISTO CHE $F=0$ SOLO SE $A=B=C=0$



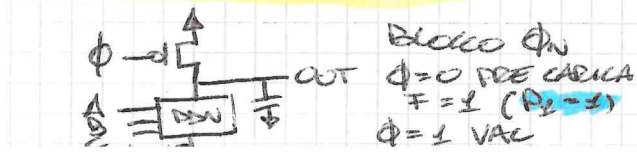
$t_p = (N-1) U e_{p0} + \frac{C_L}{U^{N-1} C_1} t_{p0} \rightarrow (N-1 \text{ BLOCCHI}) + (\text{ULTIMO BLOCCO})$

$N_{OPT} = \frac{\ln(x)}{\ln(u)} \rightarrow$ RITARDO OTTIMO: $\frac{t_p}{N} \rightarrow U_{OPT} = e$

$t_{PERT} = \ln\left(\frac{C_L}{C_{IN}}\right) e t_{p0}$



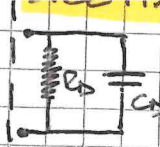
LOGICHE DINAMICHE



'LOG WP-CMOS' SERVE ϕ E $\bar{\phi}$

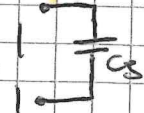
CONDIZIONE PN

- $V_D > 0 : I_D = I_S [\exp(\frac{V_D}{V_{Th}}) - 1]$
- $V_D < 0 : I_D \approx -I_S$



$R_D = \frac{V_{Th}}{I_D}$
 $C_D = \frac{\tau_{TS} I_D}{V_{Th}}$
 $= \tau_{TS} g_D$

INVERSA



$C_S = \frac{C_{S0}}{\sqrt{1 - \frac{V_D}{\phi_s}}}$

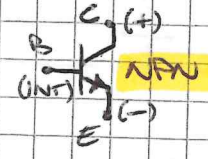
$C_D \approx 0$ e $R_D \rightarrow +\infty$

BST

NPN → ATTIVA SE $V_{BE} > 0, V_{BC} < 0$, SAT SE $V_{BE} > 0$ e $V_{BC} > 0$
 PER PNP SCAMBIO I RIFERIMENTI

SATURAZIONE: $V_{CE, SAT} = 0,1 \div 0,2V \Rightarrow I_C = \beta_F I_{BE} - (1 + \beta_F) I_{BC}$

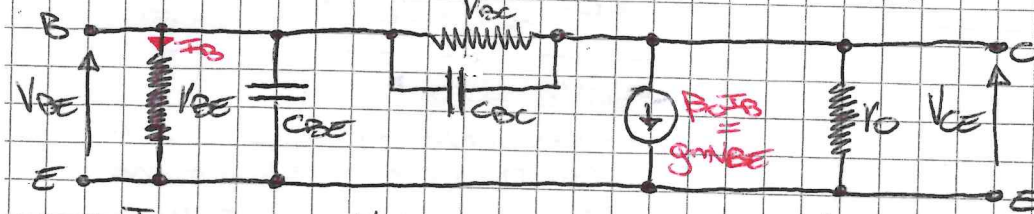
NORMALE/ATTIVA: $I_C = I_S \exp(\frac{V_{BE}}{V_{Th}})$ e $I_E = (\beta_F + 1) I_B, I_C = \beta_F I_B$



EFF. EARLY: $I_C = I_S \exp(\frac{V_{BE}}{V_{Th}}) (1 + \frac{V_{BC}}{V_A}) \approx I_S \exp(\frac{V_{BE}}{V_{Th}}) (1 + \frac{V_{CE}}{V_A})$

$\beta_F = \beta_{FO} (1 + \frac{V_{BC}}{V_A}) \approx \beta_{FO} (1 + \frac{V_{CE}}{V_A})$

IN REG. NORMALE:



$V_{CE} \rightarrow +\infty$ e $\frac{dI_C}{dV_{CE}} = g_m / \beta_F$

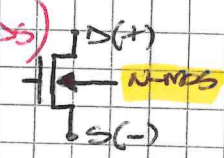
$g_m = \frac{I_C}{V_{Th}} ; V_{BE} = \frac{V_{Th}}{I_{EE}} ; r_o \approx \frac{V_A}{I_C} ; r_{ce} = \frac{V_{Th}}{I_{EC}} ; C_D = \tau_{TS} g_m ; C_S = \frac{C_{S0}}{\sqrt{1 - \frac{V_{BE}(R_E)}{\phi_s}}}$

MOSFET

QU SE $V_{GS} > V_T$ (N-MOS, DRAIN MAX TENS)

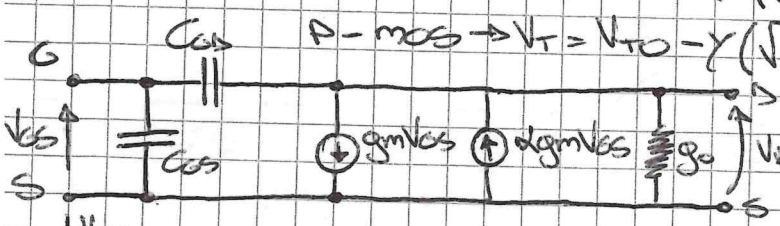
TRIODO: $V_{DS} \leq V_{GS} - |V_T| \rightarrow I_{DS} = \beta_N [(V_{GS} - |V_T|) V_{DS} - \frac{1}{2} V_{DS}^2] (1 + \lambda V_{DS})$

SAT: $V_{DS} > V_{GS} - |V_T| \rightarrow I_{DS} = \frac{\beta_N}{2} (V_{GS} - |V_T|)^2 (1 + \lambda V_{DS})$



PER P-MOS SCAMBIO I RIF E METTO IL VALORE ASSOLUTO

EFFETTO BODY: N-MOS $\rightarrow V_T = V_{T0} + \gamma (\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F})$
 P-MOS $\rightarrow V_T = V_{T0} - \gamma (\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F})$
 $\phi_F = \sqrt{k_B \cdot \ln(\frac{N_A}{n_i})}$



NOTA: EFFETTO BODY

$\lambda = \frac{\gamma}{2 \sqrt{2\phi_F + V_{SB}}}$

TRIODO: $g_m = \beta_N (1 + \lambda V_{DS}) V_{DS}$

$g_o = \beta_N (V_{GS} - |V_T| - V_{DS}) (1 + \lambda V_{DS}) + \frac{\lambda I_{DS}}{(1 + \lambda V_{DS})}$

PINCH-OFF: $g_m = \beta_N (V_{GS} - |V_T|) (1 + \lambda V_{DS})$

$g_o = \lambda \frac{\beta_N}{2} (V_{GS} - |V_T|)^2 = \frac{\lambda I_{DS}}{(1 + \lambda V_{DS})}$

AP MOSFET INTRINSECHE

OFF: $C_{GS} = C_{GD} = 0 ; C_{GB} = W L C_{ox}$

TRIODO: $C_{GB} = 0 ; C_{GS} = C_{GD} = \frac{1}{2} W L C_{ox}$

SAT: $C_{GS} = C_{GB} = 0 ; C_{GD} = \frac{2}{3} W L C_{ox}$

PARASSITE

$C_{GDP} = X_d C_{ox} + C_{SV}$

$C_G \approx W L C_{ox} + 2 W C_{GDP}$

SRIDA h

$I = h_{11} I_i + h_{12} V_o$

$O = h_{21} I_i + h_{22} V_o$

IRIDA g

$I_i = g_{11} V_i + g_{12} I_o$

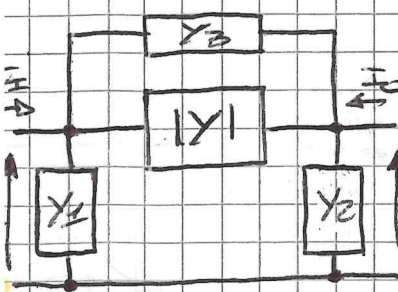
$V_o = g_{21} V_i + g_{22} I_o$

AMMETTENZA

$\|Y\| = \begin{vmatrix} Y_i & Y_e \\ Y_f & Y_o \end{vmatrix}$

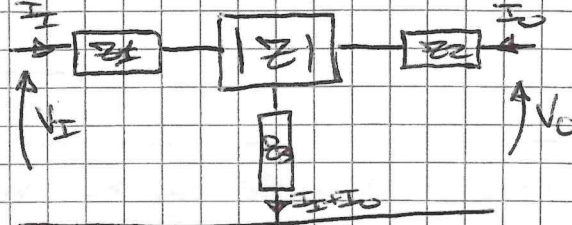
MAT SINGOLARE
 $\hookrightarrow \det = 0$
 $\hookrightarrow$ NON INVERTIBILE

PROP. MAT. AMMETTENZA



- Y_1 SOMMA A Y_3
- Y_2 SOMMA A Y_4
- Y_3 SOMMA DIAG. PRINCIPALE E SOTTRAGGO SULLA SECONDARIA

PROP. MAT. IMPEDENZA



- Z_1 SOMMA A Z_3
- Z_2 SOMMA A Z_4
- Z_3 SOMMA A TUTTE LE COMPONENTI

	z	y	h
A_V	$\frac{z_f z_c}{D_o + z_i z_c}$	$-\frac{y_f}{y_o + y_c}$	$-\frac{h_f}{D_o + h_i y_c}$
A_I	$-\frac{z_f}{z_o + z_c}$	$\frac{y_f y_c}{D_o + y_i y_c}$	$\frac{h_f}{1 + h_o z_c}$
$Z_i \text{ o } Y_i = \frac{1}{Z_i}$	$Z_i = z_i - \frac{z_f z_c}{z_o + z_c}$	$Y_i = y_i - \frac{y_f y_c}{y_o + y_c}$	$Z_i = h_i - \frac{h_o h_f}{h_o + y_c}$
$Z_o \text{ o } Y_o = \frac{1}{Z_o}$	$Z_o = z_o - \frac{z_f z_c}{z_i + z_c}$	$Y_o = y_o - \frac{y_f y_c}{y_i + y_c}$	$Y_o = h_o - \frac{h_i h_f}{h_i + z_c}$
$\frac{V_{aco}}{V_o}$	$\frac{z_f}{z_i + z_c}$	$-\frac{y_f}{D_o + y_o + y_c}$	$-\frac{h_f}{D_o + h_o z_c}$
$\frac{I_{acc}}{I_o}$	$-\frac{z_f}{D_o + y_o + z_c}$	$\frac{y_f}{y_i + y_c}$	$\frac{h_f}{h_i y_c + 1}$

FUNZIONI DI RETE

$A_V = \frac{V_o}{V_i}$; $A_I = \frac{I_o}{I_i}$; $Z_{IN} = \frac{V_i}{I_i}$

$Z_o = \frac{V_o}{I_o} |_{I_i=0} \rightarrow$ SEGNALE D'INGRESSO

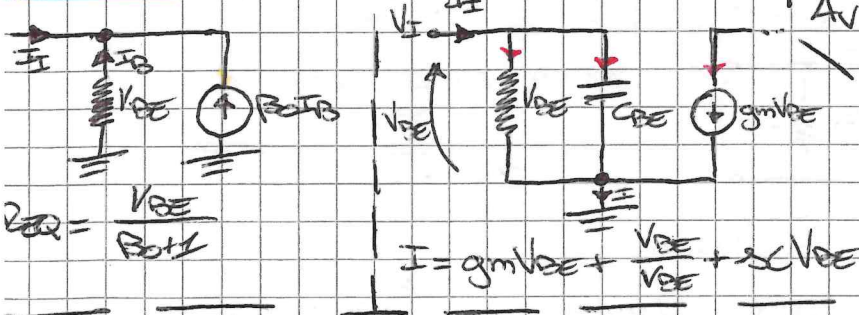
ALTRE COSE

- MOS CON. A DIODO STA AI LIMITI DELLA SATURAZIONE
- BJT CON. A DIODO STA SEMPRE IN NORM/ATTIVA
 $V_{BE} = V_{CE} \rightarrow I_E = I_C + \frac{I_C}{\beta_F}$
- FORMA DI BODE: CALCOLO I TERMINI CHE NON DIP. DA S
 $(S+P) = P(1+\frac{S}{P})$
- FUNZ DI RETE EASY:

$A_V = \frac{Y_{21}}{Y_{22}}$ e $A_I = \frac{Y_{21}}{Y_{11}}$

$A_V = \frac{Z_{21}}{Z_{11}}$ e $A_I = \frac{Z_{21}}{Z_{22}}$

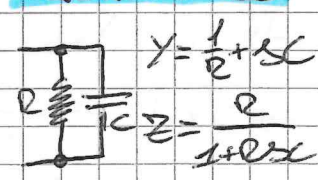
ESEMPLI



PARTITORE DI CORRENTE



EQUIVALENZE



$Y = \frac{1}{R} + sC$
 $Z = R + \frac{1}{sC}$

$V_x = (R_D + R_S) I_D (R_1 || R_2 || R_3 ...)$

$R_1 || R_2 = \frac{R_1 \cdot R_2}{R_1 + R_2}$